

Mecklenburg County Sample — Shopping Center / Retail



Traffic Impact Study

Prepared August 8, 2026 · 35.1954, -80.7643

SIS sample draft — full tier

Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Analysis (TIA) evaluates the anticipated traffic impacts of the proposed Mecklenburg County Sample — Shopping Center / Retail within Charlotte MSA, North Carolina (City of Charlotte / Mecklenburg County). The analysis follows the structure of the NCDOT Policy on Street and Driveway Access to North Carolina Highways (July 2003) Chapter 5 and the NCDOT Congestion Management Capacity Analysis Guidelines (Standards, March 2022; Best Practices, November 20, 2024). 4 study intersections within a 0.50-mile radius are analyzed under Existing, No-Build, and Build conditions.

4	0	2	6,800
INTERSECTIONS	LOS DROPS	AT LOS E/F	DAILY TRIPS (RAW)

1.0 INTRODUCTION

Project name	Mecklenburg County Sample — Shopping Center / Retail
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	35.1954°, -80.7643°
Build-out (horizon) year	2027
Controlling jurisdiction	City of Charlotte / Mecklenburg County
MPO	CRTPO (Charlotte Regional Transportation Planning Organization) — 2026-2035 TIP
Programmed projects source	NCDOT 2026-2035 STIP (adopted July 2025), as amended

1.1 TIA Applicability

Charlotte UDO Article 32, Section 32.1 (effective 6/1/2023) requires a Comprehensive Transportation Review (CTR) — Multimodal Assessment + TDM Assessment + Traffic Impact Study — when Charlotte Streets Manual Chapter 3, Table 3.1 thresholds are met (reported secondary-source values: >1,500 vpd or >150 peak-hour trips in lower-intensity districts; >2,000 vpd or >200 peak-hour trips in medium/high-intensity districts — confirm against the current Table 3.1 at scoping). The CTR applies at permitting even for by-right development.

The proposed development's estimated raw trip generation of 6,800 vehicles per day meets or exceeds the 3,000 vpd threshold at which NCDOT may require a TIA (no reductions are applied in the threshold test).

1.2 Scoping

CDOT publishes a Rezoning Transportation Analysis memo per rezoning petition; the TIS must be prepared by a North Carolina licensed Professional Engineer. NCDOT coordination applies on state-system streets. The background growth factor and the list of approved background developments must be agreed in the Approved Scoping Document prior to TIA submittal; a placeholder for that document is carried in the Appendix.

2.0 METHODOLOGY

NCDOT requires HCM-based analysis; the Department currently utilizes Synchro 11 (with mandatory SimTraffic simulation, minimum ten runs), HCS for freeway facilities, and Sidra for roundabouts. This screening analysis applies HCM-consistent computation; Synchro 11-compatible models and SimTraffic outputs are to be provided at formal submittal.

Peak periods	AM and PM weekday peak hours (minimum); counts Tue–Thu, school in session, less than 12 months old
PHF (future conditions)	0.90 per Standards
Heavy vehicles	Average of duals and TTSTs, 2% minimum
RTOR (future analysis)	Not permitted per Standards
Turn-lane storage	Greater of Synchro 95th percentile and SimTraffic maximum queue, rounded up to 25 ft; 100 ft minimum
Roundabout threshold	v/c 0.85
Study-area rule	Intersections where site trips add 10% or more to background volumes on any approach/movement

2.1 Trip Generation Basis

NCDOT's Best Practices call for the ITE Trip Generation Manual with the Department's Rate-vs-Equation spreadsheet, internal capture per the NCHRP 684 spreadsheet method (vehicle occupancy 1.1, 4,000-ft maximum walking distance), and pass-by limited to retail uses with multi-use pass-by capped at 10% of the adjacent street volume. This screening analysis applies public-data trip-generation rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), with every rate tagged so the jurisdiction-approved ITE figure can be substituted at submittal; applied internal-capture and pass-by credits must be reconciled against the NCDOT limits at scoping.

3.0 ANALYSIS CONDITIONS AND MEASURES OF EFFECTIVENESS

Scenarios follow the NCDOT Standards set: Existing Base Year; No-Build Design Year (build-out); and Design Year Build. Where a scoped project is phased, intermediate build-out years are analyzed with earlier-phase trips carried as site trips (never background). Where an impacted STIP project is in planning, design, or construction (or within 5 years post-construction) and site traffic was not in the STIP forecast, an additional STIP design-year analysis is coordinated per the Best Practices TIA/STIP flow chart.

3.1 Intersection MOE Summary (NCDOT format)

Approaches are ordered EB, WB, NB, SB and movements left-through-right per the Best Practices reporting format. Signalized intersections report control delay and LOS overall and per lane group (any v/c > 1.0 is reported as LOS F regardless of delay); unsignalized intersections report no overall LOS — results are per lane group with the conflicting movement. SimTraffic maximum-queue columns are populated at submittal.

No.	Intersection	Existing LOS/ Delay	No-Build LOS/ Delay	Build LOS/Delay
1	North Sharon Amity Road & Holbrook Drive	B / 14.6s	B / 14.6s	B / 17.1s
2	Near North Sharon Amity Road	F / 120.0s	F / 120.0s	F / 120.0s
3	North Sharon Amity Road & Charleston Drive	B / 16.3s	B / 16.3s	B / 17.6s
4	Albemarle Road & North Sharon Amity Road	F / 120.0s	F / 120.0s	F / 120.0s

4.0 MITIGATION CRITERIA CHECK (POLICY CH. 5.J)

NCDOT requires improvements when, comparing base-network and project conditions: total average delay increases by 25% or more while remaining at the same LOS; the LOS degrades by one level; or the intersection operates at LOS F. Turn-lane mitigation applies where the 95th-percentile queue exceeds existing storage. Signal-timing changes alone are not considered mitigation. The District Engineer makes the final mitigation determination.

No.	Intersection	Δ delay	LOS drop	LOS F	Ch. 5.J result
1	North Sharon Amity Road & Holbrook Drive	+17%	no	no	none indicated
2	Near North Sharon Amity Road	+0%	no	YES	MITIGATION INDICATED
3	North Sharon Amity Road & Charleston Drive	+8%	no	no	none indicated

No.	Intersection	Δ delay	LOS drop	LOS F	Ch. 5.J result
4	Albemarle Road & North Sharon Amity Road	+0%	no	YES	MITIGATION INDICATED

5.0 CONCLUSIONS AND SUBMITTAL NOTES

0 intersections project a LOS drop and 2 operate at LOS E or F under Build conditions; mitigation per Policy Ch. 5.J should be developed with the District Engineer.

Data conventions: NCDOT forecasts supply AADT, % duals and TTSTs, directional split (D), and design-hour factor (K); forecast conversion to peak-hour volumes uses the Intersection Analysis Utility (IAU) in the NCDOT Traffic Engineering Suite. AADT reference: NCDOT ArcGIS AADT mapping application. Crash analysis: a site-radius crash assessment from NCDOT's TEAAS (Traffic Engineering Accident Analysis System) is prepared at submittal — not generated by this screening tool.

This TIA is to be prepared under the direct charge of and sealed by a licensed North Carolina Professional Engineer with expertise in traffic engineering (Policy Ch. 5.A; 21 NCAC 56 .1103 — seal with signature over or adjacent to the seal and date of signing; firm name, address, and NC firm license number on the title block). TIAs marked confidential remain so until a driveway permit is requested or the project is publicly announced.

SCREENING DISCLOSURE: this document is a screening-level draft. It does not include the Approved Scoping Document, SimTraffic outputs, Synchro models, TEAAS crash analysis, or signed/sealed certification — all required for a formal NCDOT TIA submittal.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: $a=28507$, $b=-0.02$, $c=-0.123$) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
North Sharon Amity Road & Holbrook Drive	0.02	0.1	401	48.3%
Near North Sharon Amity Road	0.14	0.3	217	26.1%
North Sharon Amity Road & Charleston Drive	0.19	0.5	172	20.7%
Albemarle Road & North Sharon Amity Road	0.47	1.1	41	4.9%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 92% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 8% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 2.00. Per-signal v/c , delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

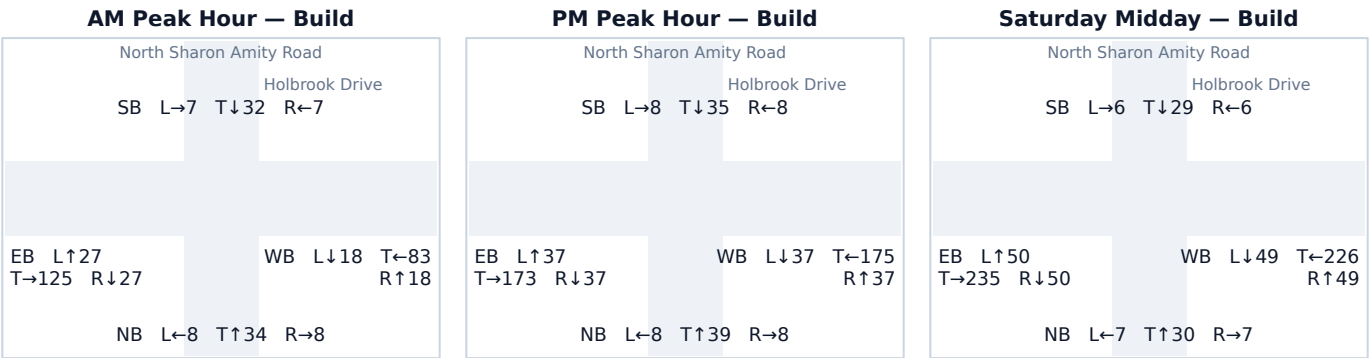
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 North Sharon Amity Road & Holbrook Drive

Signal ID	charlotte-172660814
Location	NPA 16 · 0.02 mi from site
Added PM peak trips	401
Existing / No-Build (PM)	LOS B · 14.6 s/veh · v/c 0.11
Build (PM)	LOS B · 17.1 s/veh · v/c 0.33
Δ control delay	+2.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	55	0	55	0.07 → 0.07	14.2 → 14.2	B → B	32
SB	51	0	51	0.06 → 0.06	14.2 → 14.2	B → B	30
EB	55	192	247	0.07 → 0.31	14.2 → 16.8	B → B	163
WB	40	209	249	0.05 → 0.31	14.0 → 16.8	B → B	164

Affected movements (PM peak project trips)

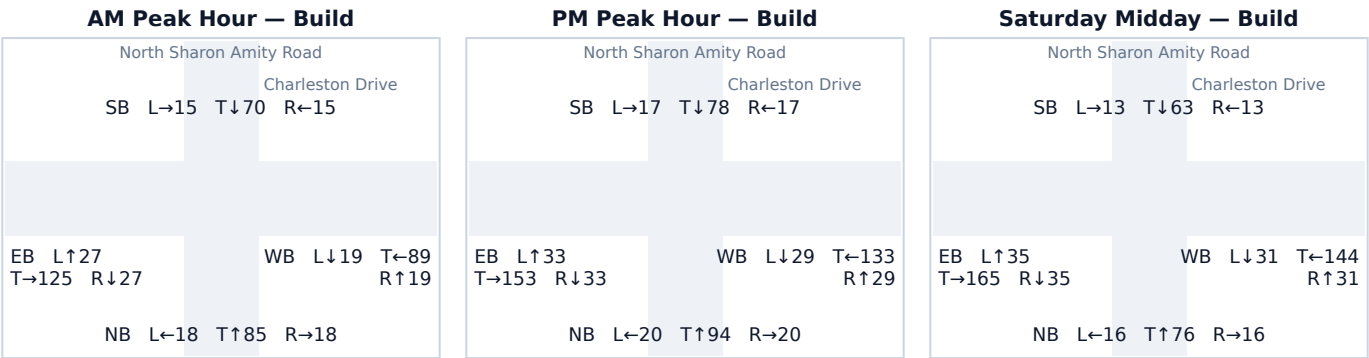
Movement	Project trips	% of project trips
WB Through	209	52%
EB Through	192	48%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 North Sharon Amity Road & Charleston Drive

Signal ID	charlotte-7810838990
Location	NPA 52 · 0.19 mi from site
Added PM peak trips	172
Existing / No-Build (PM)	LOS B · 16.3 s/veh · v/c 0.27
Build (PM)	LOS B · 17.6 s/veh · v/c 0.36
Δ control delay	+1.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	134	0	134	0.17 → 0.17	15.2 → 15.2	B → B	82
SB	112	0	112	0.14 → 0.14	14.9 → 14.9	B → B	68
EB	137	83	219	0.17 → 0.27	15.2 → 16.3	B → B	142
WB	101	89	191	0.13 → 0.24	14.7 → 15.9	B → B	121

Affected movements (PM peak project trips)

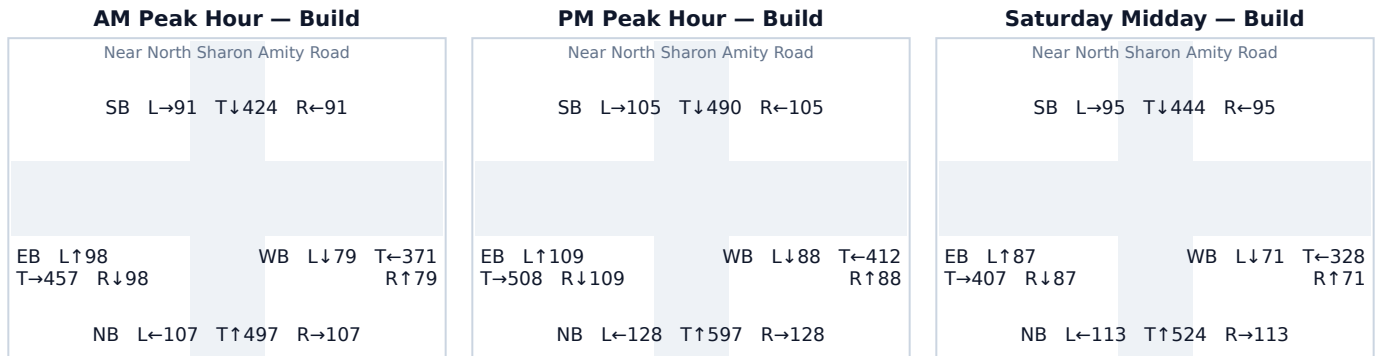
Movement	Project trips	% of project trips
WB Through	89	52%
EB Through	83	48%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Near North Sharon Amity Road

Signal ID	charlotte-2181076118
Location	NPA 16 · 0.14 mi from site
Added PM peak trips	217
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.47
Build (PM)	LOS F · 120.0 s/veh · v/c 1.59
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	740	113	853	0.91 → 1.05	39.6 → 71.1	D → E	873
SB	595	104	700	0.74 → 0.86	26.2 → 34.1	C → C	649
EB	726	0	726	0.90 → 0.90	37.4 → 37.4	D → D	690
WB	588	0	588	0.73 → 0.73	25.8 → 25.8	C → C	495

Affected movements (PM peak project trips)

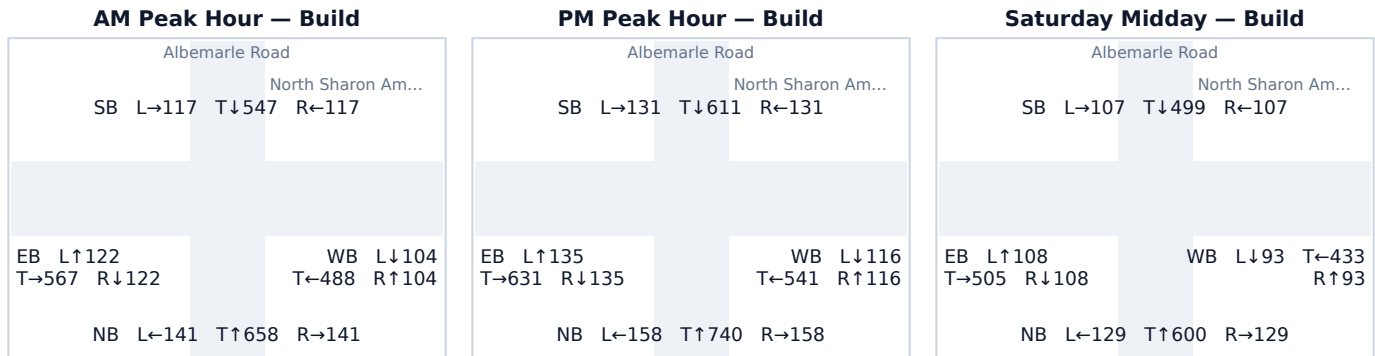
Movement	Project trips	% of project trips
NB Through	113	52%
SB Through	104	48%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Albemarle Road & North Sharon Amity Road

Signal ID	charlotte-2181076171
Location	NPA 16 · 0.47 mi from site
Added PM peak trips	41
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.98
Build (PM)	LOS F · 120.0 s/veh · v/c 2.00
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	1,035	21	1,056	1.28 → 1.30	120.0 → 120.0	F → F	1,081
SB	854	20	873	1.05 → 1.08	71.3 → 79.3	E → E	893
EB	901	0	901	1.11 → 1.11	91.6 → 91.6	F → F	922
WB	773	0	773	0.95 → 0.95	46.2 → 46.2	D → D	769

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	21	51%
SB Through	20	49%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.